

FINAL PROJECT

REPORT

DSA 210- MELİS TUNAR

MOTIVATION:

The role of digital devices increased and how increasing screen time impacts productivity and well-being is a question many people ask. In my personal experience, I was curious about if my screen time affected my sleep quality, focus or stress-level. This motivated me to collect the daily data and analyze if there is a noticeable relationship between the factors. Based on the findings, this project aims to adjust personal daily routines in ways that enhance focus, well-being and productivity while reducing stress.

DATA SOURCE AND COLLECTION:

The dataset was self-collected for 28 days through daily manual input. Each day, data was collected from phone or by daily collection.

1. Measured Variables:

- Total screen time (in hours, collected from iphone)
- Evening screen time (in hours, between 20.00 and 00.00, collected from iphone)
- Time spent on social media (in hours, collected from iphone)
- Sleep duration (in hours)
- Mood, stress and focus (collected daily on scale 1-5)

2. Derived Variables:

- Evening screen ratio: Derived from the ratio of evening screen time and total screen time
- Social media ratio: Derived from the ratio of time spent on social media and total screen time
- Productivity score: A weighted formula was implemented to find a productivity score which is;

$$Productivity = (Focus * 0.5) + (Sleep * 0.3) + (Mood * 0.2)$$

METHODOLOGY:

The analysis contains several stages: Data cleaning, Exploratory Data Analysis (EDA), Hypothesis Testing and applying Machine Learning Method.

- 1. Data Cleaning:** The dataset is checked for missing values and formatting issues. Derived variables were computed such as productivity score, evening screen ratio and social media ratio.
- 2. Exploratory Data Analysis:** Summary statistics (mean, median, standard deviation) were computed to understand the distribution of variables and identify potential relationship. Distributions are visualized using histograms and scatterplots. These were used to gain understanding the correlation between variables. Then, correlation analysis applied to test the hypotheses.
- 3. Hypothesis Testing:** There are 4 hypotheses. Pearson correlation coefficient was used to test the linear relationship between variables and significance level $\alpha=0.05$ is used to decide to reject or fail to reject the Null Hypotheses.
- 4. Machine Learning Methods:** In addition to statistical analysis, a supervised machine learning method was applied. A Multiple Linear Regression model was used to predict daily productivity score based on behavioral features. The dataset was divided using a standard train-test split to evaluate the model's performance on unseen data. Due to the limited dataset size, the goal was not to have a high predictive accuracy system, but rather to understand how daily screen usage impacts productivity. The model was evaluated using

Mean Squared Error (MSE) and R^2 score to interpret how productivity is influenced by daily digital usage.

FINDINGS AND INTERPRETATION:

Four main hypotheses were tested:

H1: Evening screen time – Sleep: No significant relationship is found between evening screen time and sleep, so we Fail to Reject Null Hypothesis (H0).

H2: Sleep – Productivity Score: Higher sleep hours didn't significantly result in a higher productivity score, so we Fail to Reject Null Hypothesis (H0).

H3: Evening Screen Ratio – Next Day Focus: The ratio of evening screen time didn't significantly change the next day focus, so we Fail to Reject Null Hypothesis (H0).

H4: Social Media Usage – Stress: Time spent on social media did not show a statistically significant correlation with stress, so we Fail to Reject Null Hypothesis (H0).

The reason why no significant relationship observed may be influenced by the small dataset size. The observations were made for 28-days, increasing the observation period may provide more accurate results.

MACHINE LEARNING APPLICATIONS:

In addition to simple correlations, Multiple Linear Regression model was used.

The goal was to predict the Productivity Score based on sleep, focus and screen ratios.

The data was split into training and testing sets. The model was evaluated using Mean Squared Error (MSE) and R^2 score. While The dataset size was limited, the coefficients indicated that factors like focus and sleep had the strongest impact on determining the productivity score.

LIMITATIONS AND FUTURE WORK:

The dataset consists of the observations over 28-day data, which limits statistical power and ability to detect to find significant relationships. Also the data was collected from a single individual.

Also, some data was collected manually, which may cause some subjectivity and bias.

Future work can involve extending the dataset collection period and may include more data that could affect the relationship. With a larger dataset, the results may give more accurate understanding about the correlations.

CONCLUSION:

This project applied EDA, Hypothesis testing and Machine Learning method to explore the relationship between daily digital usage, well-being and daily productivity using the daily collected personal data. Even though the results show no statistically significant correlation between the variables, the analysis provides a structured framework for understanding the behavioral data. The findings can be used for making improvements to daily routines for better productivity, focus and well-being while reducing stress.